Do Fake News Detectors Generalize Across Datasets?

Experimental results: cross-dataset evaluation of classical and deep-learning models

1. Summary

The original Fake News Detector reached about 97% accuracy on one dataset (ISOT). This study asks what happens when models trained on one news dataset are tested on another. Across classical machine learning (TF-IDF + LinearSVC, logistic regression, Naive Bayes) and four deep-learning models (DistilBERT, BiLSTM, TextCNN and a small transformer), accuracy was **about 90-99% on the news dataset a model was trained on but only 56-67% on a different news dataset** in the cleaned runs, close to the 55-57% a model would get by always guessing the majority class. Higher cross-dataset values (72-86%) appeared only when formatting cues were left in the text.

- Cross-dataset accuracy collapsed for every model family, which suggests the problem lies in what the datasets let models learn and not in the choice of architecture.
- ISOT turned out to be almost entirely contained inside WELFake: 38,261 of WELFake's 61,793 articles (61.9%) are copies of ISOT articles. Testing across these two datasets without removing the copies gives misleading numbers.
- An apparently good transfer result (DistilBERT 85.5% from WELFake to ISOT) fell to 67.2% once all punctuation and formatting cues were removed, which suggests that part of it came from formatting leakage.
- LIAR (short political claims) is not learnable by article-style models: every model scores near chance on it.

2. Setup

Datasets (after removing articles shared between datasets):

Dataset	Articles	% labelled true	Notes
ISOT	38,261	55.4	Reuters (true) vs unreliable sites (fake); 2016-2017 US politics
WELFake-novel	23,532	56.9	WELFake minus every article that is also in ISOT
LIAR (binary)	12,827	72.3	One-sentence political claims; lenient binary labelling

The 'always say true' baseline is therefore 55.4% (ISOT), 56.9% (WELFake-novel) and 72.3% (LIAR). Accuracy above these values is the real signal.

- **Protocol:** train on 80% of one dataset, test on the held-out 20% (in-domain) and on the full other datasets (cross-dataset). Shared articles were removed from all datasets before any split.
- **Label check:** the WELFake copy used had its labels inverted relative to ISOT. This was detected because the duplicated articles carried opposite labels in the two files (0% agreement), and corrected before all experiments reported here (100% agreement afterwards).
- **Text cleaning modes:** *raw* (as written), *delaked* (URLs and wire-service terms removed, punctuation kept) and *strict* (also the 'CITY (Reuters) -' dateline and all punctuation removed; letters and digits only).

- **Metrics:** accuracy, and balanced accuracy (50% = constant guess). Mean ± standard deviation over seeds where stated.

3. Classical models (TF-IDF + LinearSVC)

Final cleaning pipeline from the original project; full data, 5 seeds. Accuracy % (rows = trained on, columns = tested on).

Trained on \ Tested on	ISOT	WELFake-novel	LIAR
ISOT	97.5 ± 0.1	56.6 ± 0.2	32.0 ± 0.3
WELFake-novel	66.6 ± 0.4	92.7 ± 0.1	44.6 ± 1.5
LIAR	56.5 ± 0.5	57.6 ± 0.6	70.3 ± 0.8

Highlighted: the two cross-dataset cells between the news-article datasets. Logistic regression and Naive Bayes show the same pattern. With basic cleaning (no leakage removal) ISOT in-domain accuracy is 99.3%, reproducing the suspiciously high score the original project started from.

The ISOT-trained model labels only about 25% of WELFake-novel articles as true when 57% are true: it treats real news from other outlets as fake, consistent with ISOT's true class being almost entirely Reuters copy.

4. Deep-learning models

4.1 Strict text cleaning, 3 seeds (accuracy % ± std; balanced accuracy in brackets)

DistilBERT	ISOT	WELFake-novel	LIAR
Trained on ISOT	98.8 ± 0.2 (98.6)	63.8 ± 1.1 (67.3)	32.2 ± 1.1 (50.3)
Trained on WELFake-novel	67.2 ± 1.0 (63.6)	96.5 ± 0.1 (96.2)	46.7 ± 7.1 (52.9)
Trained on LIAR	55.2 ± 0.6 (50.1)	58.8 ± 0.3 (52.3)	73.8 ± 0.5 (59.0)

Compact transformer	ISOT	WELFake-novel	LIAR
Trained on ISOT	95.0 ± 0.4 (94.9)	55.8 ± 1.3 (59.4)	44.0 ± 3.4 (50.6)
Trained on WELFake-novel	62.0 ± 0.3 (58.7)	89.9 ± 0.7 (89.1)	32.3 ± 1.2 (51.0)
Trained on LIAR	55.0 ± 0.3 (49.8)	57.6 ± 0.4 (50.8)	72.1 ± 0.3 (56.5)

The compact transformer is a small encoder trained from scratch, inspired by the reference paper's design. It is not a replication of that paper's model.

4.2 Effect of text cleaning (accuracy %, news-article pair only)

Model / text mode	ISOT in-domain	WELFake-novel in-domain	ISOT → WELFake-novel	WELFake-novel → ISOT
DistilBERT, raw	99.96	98.09	56.7	77.0
DistilBERT, delaked	99.97	97.83	55.0	85.5
DistilBERT, strict (3 seeds)	98.8	96.5	63.8	67.2
Compact transf., raw	99.9	93.0	60.2	62.7
Compact transf., delaked	99.0	92.8	60.1	60.9
Compact transf., strict (3 seeds)	95.0	89.9	55.8	62.0

Raw and delaked rows are single-seed runs; strict rows are 3-seed means.

4.3 Other architectures (single seed, raw / delaked)

Model	ISOT in-domain	WELFake-novel in-domain	ISOT → WELFake-novel	WELFake-novel → ISOT
BiLSTM	99.9 / 99.8	94.9 / 92.6	58.2 / 61.3	47.4 / 58.2
TextCNN	99.9 / 99.9	95.7 / 95.8	46.9 / 49.9	58.7 / 57.4

Seed-to-seed variation was not measured for these runs (it was 0.2-1.3 points in the news-pair cells of the 3-seed models), so differences between the raw and delaked values, such as the BiLSTM's 47.4 vs 58.2, are unconfirmed.

5. Findings

- **Generalization fails across architectures.** From a 4M-parameter CNN to a 67M pretrained transformer, in-domain accuracy on the news datasets is about 90-100%, while cross-dataset accuracy between them is 56-67% in the cleaned runs (classical delaked and deep-learning strict). Under strict cleaning the in-domain vs cross-dataset gap averages 32 points for DistilBERT and 34 for the compact transformer.
- **Formatting cues inflated one apparent success.** With punctuation left in, DistilBERT trained on WELFake-novel scored 85.5% on ISOT. With all punctuation removed it scored 67.2% (±1.0 over 3 seeds). A plausible explanation is that the model learned punctuation and dateline conventions from Reuters-style articles inside WELFake-novel that happen to match ISOT; this was not tested directly.
- **The apparent asymmetry mostly disappears.** Earlier runs suggested that training on WELFake transfers much better to ISOT than the reverse (77-85% vs 55-57%). Under strict cleaning the two directions differ by only 3-6 points for the two transformers.
- **Pretraining helps modestly.** Under strict cleaning DistilBERT beats the from-scratch compact transformer in the cross-dataset cells (63.8 vs 55.8 and 67.2 vs 62.0, a 5-8 point advantage). That is small relative to the 30-point gap from in-domain accuracy.
- **Removing punctuation costs little in-domain.** DistilBERT still reaches 98.8% on ISOT using only lowercase letters and digits, so articles in a single dataset are separable by vocabulary and style alone. That signal does not carry over to other datasets.
- **LIAR is not a fair test for these models.** They were trained on article-length text; LIAR's median claim has about 20 words. Models trained on ISOT or WELFake predict 'fake' for most claims, and LIAR-trained models predict 'true' for most articles.

6. Limitations

- Duplicates were found by exact match after cleaning, so near-duplicates may remain. Even the cross-dataset numbers could therefore be slightly optimistic.
- 'Fake' is defined differently in each dataset, and WELFake merges several sources. Part of the gap may be label inconsistency and not model failure.
- Seeds change the training split and initialisation only. The cross-dataset test sets are fixed (capped at 20,000 articles), so the reported standard deviations do not reflect test-set sampling.
- Raw and delaked deep-learning runs use one seed; classical results use five; strict deep-learning results use three.
- The LIAR file is a binary version where only the two lowest truthfulness ratings count as false. It is much more skewed (72% true) than the original six-way labels suggest.
- These results do **not** test the reference paper's own model or numbers. The paper's architecture was not replicated here; the compact transformer is only inspired by it.
- All data are English-language news, largely 2016-2017 US politics.

7. Suggested next steps

- Add more seeds (at least 5) for DistilBERT and report confidence intervals; repeat the strict-mode comparison for BiLSTM and TextCNN.
- Inspect which words each model relies on (for example LIME or top TF-IDF features) to test the explanations offered above rather than assume them.
- Add a third news-article dataset and an AI-generated fake-news test set to see whether the failure to transfer extends beyond ISOT and WELFake.
- If a claim about the reference paper is wanted, reimplement its model from its published specification and repeat the same cross-dataset protocol.